

**2013**  
**HIGHER SCHOOL CERTIFICATE  
EXAMINATION**

# Mathematics Extension 1

## General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen  
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

## Total marks – 70

### **Section I** Pages 2–7

#### **10 marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

### **Section II** Pages 8–15

#### **60 marks**

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

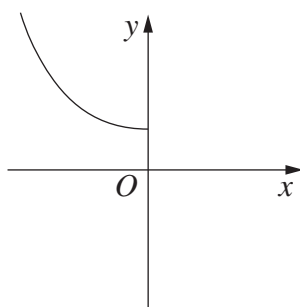
Use the multiple-choice answer sheet for Questions 1–10.

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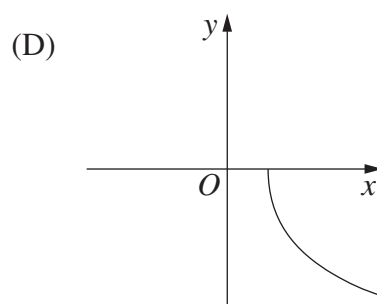
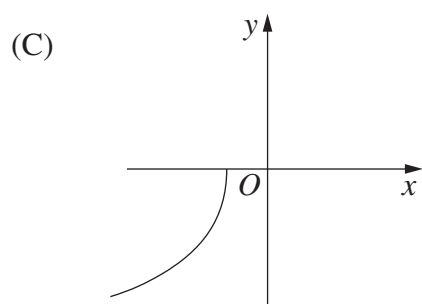
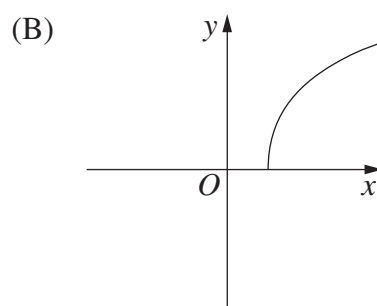
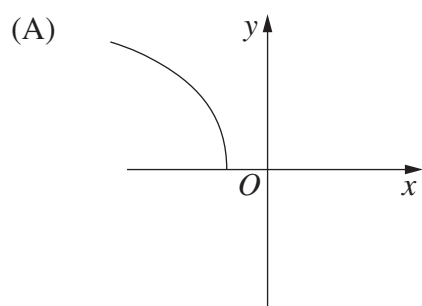
- 1 The polynomial  $P(x) = x^3 - 4x^2 - 6x + k$  has a factor  $x - 2$ .

What is the value of  $k$ ?

- (A) 2  
(B) 12  
(C) 20  
(D) 36
- 2 The diagram shows the graph  $y = f(x)$ .

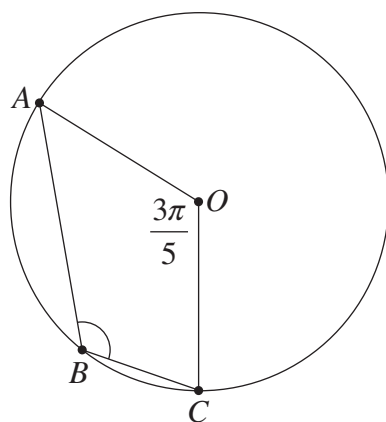


Which diagram shows the graph  $y = f^{-1}(x)$ ?



- 3 The points  $A$ ,  $B$  and  $C$  lie on a circle with centre  $O$ , as shown in the diagram.

The size of  $\angle AOC$  is  $\frac{3\pi}{5}$  radians.

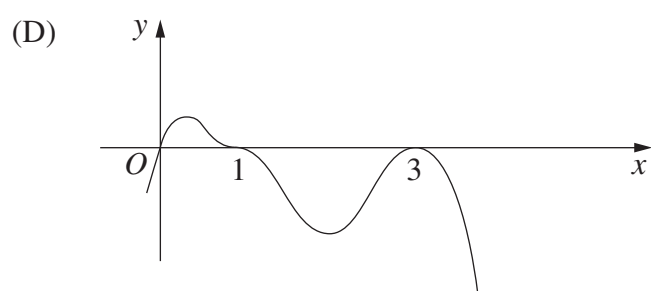
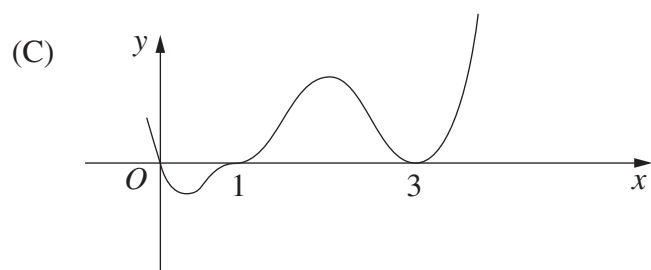
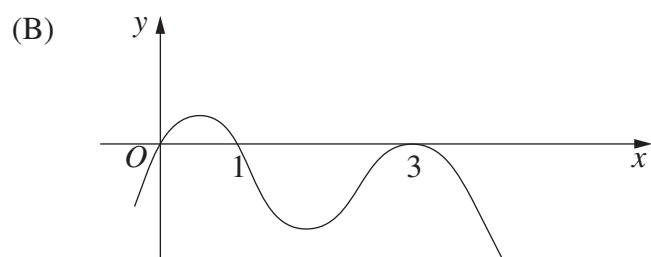
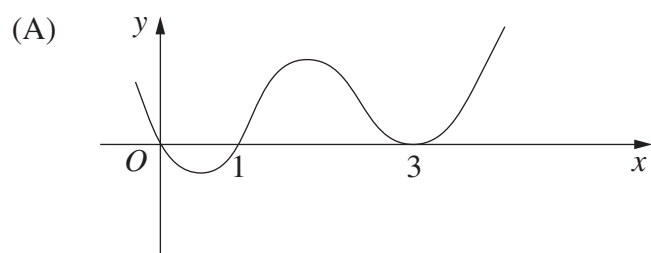


NOT TO  
SCALE

What is the size of  $\angle ABC$  in radians?

- (A)  $\frac{3\pi}{10}$
- (B)  $\frac{2\pi}{5}$
- (C)  $\frac{7\pi}{10}$
- (D)  $\frac{4\pi}{5}$

4 Which diagram best represents the graph  $y = x(1-x)^3(3-x)^2$ ?



5 Which integral is obtained when the substitution  $u = 1 + 2x$  is applied to  $\int x\sqrt{1 + 2x} \, dx$ ?

(A)  $\frac{1}{4} \int (u - 1)\sqrt{u} \, du$

(B)  $\frac{1}{2} \int (u - 1)\sqrt{u} \, du$

(C)  $\int (u - 1)\sqrt{u} \, du$

(D)  $2 \int (u - 1)\sqrt{u} \, du$

6 Let  $|a| \leq 1$ . What is the general solution of  $\sin 2x = a$ ?

(A)  $x = n\pi + (-1)^n \frac{\sin^{-1} a}{2}$ ,  $n$  is an integer

(B)  $x = \frac{n\pi + (-1)^n \sin^{-1} a}{2}$ ,  $n$  is an integer

(C)  $x = 2n\pi \pm \frac{\sin^{-1} a}{2}$ ,  $n$  is an integer

(D)  $x = \frac{2n\pi \pm \sin^{-1} a}{2}$ ,  $n$  is an integer

- 7 A family of eight is seated randomly around a circular table.

What is the probability that the two youngest members of the family sit together?

(A)  $\frac{6!2!}{7!}$

(B)  $\frac{6!}{7!2!}$

(C)  $\frac{6!2!}{8!}$

(D)  $\frac{6!}{8!2!}$

- 8 The angle  $\theta$  satisfies  $\sin \theta = \frac{5}{13}$  and  $\frac{\pi}{2} < \theta < \pi$ .

What is the value of  $\sin 2\theta$ ?

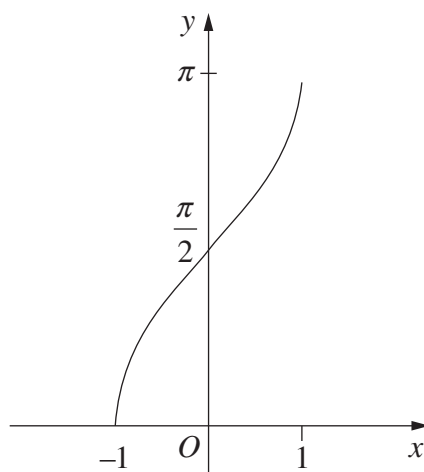
(A)  $\frac{10}{13}$

(B)  $-\frac{10}{13}$

(C)  $\frac{120}{169}$

(D)  $-\frac{120}{169}$

- 9 The diagram shows the graph of a function.



Which function does the graph represent?

- (A)  $y = \cos^{-1} x$
- (B)  $y = \frac{\pi}{2} + \sin^{-1} x$
- (C)  $y = -\cos^{-1} x$
- (D)  $y = -\frac{\pi}{2} - \sin^{-1} x$
- 10 Which inequality has the same solution as  $|x + 2| + |x - 3| = 5$ ?
- (A)  $\frac{5}{3 - x} \geq 1$
- (B)  $\frac{1}{x - 3} - \frac{1}{x + 2} \leq 0$
- (C)  $x^2 - x - 6 \leq 0$
- (D)  $|2x - 1| \geq 5$

## Section II

**60 marks**

**Attempt Questions 11–14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet.

- (a) The polynomial equation  $2x^3 - 3x^2 - 11x + 7 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ . **1**

Find  $\alpha\beta\gamma$ .

- (b) Find  $\int \frac{1}{\sqrt{49 - 4x^2}} dx$ . **2**

- (c) An examination has 10 multiple-choice questions, each with 4 options. In each question, only one option is correct. For each question a student chooses one option at random. **2**

Write an expression for the probability that the student chooses the correct option for exactly 7 questions.

- (d) Consider the function  $f(x) = \frac{x}{4 - x^2}$ .
- (i) Show that  $f'(x) > 0$  for all  $x$  in the domain of  $f(x)$ . **2**
- (ii) Sketch the graph  $y = f(x)$ , showing all asymptotes. **2**

**Question 11 continues on page 9**



Question 11 (continued)

(e) Find  $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{3x}$ . **1**

(f) Use the substitution  $u = e^{3x}$  to evaluate  $\int_0^{\frac{1}{3}} \frac{e^{3x}}{e^{6x} + 1} dx$ . **3**

(g) Differentiate  $x^2 \sin^{-1} 5x$ . **2**

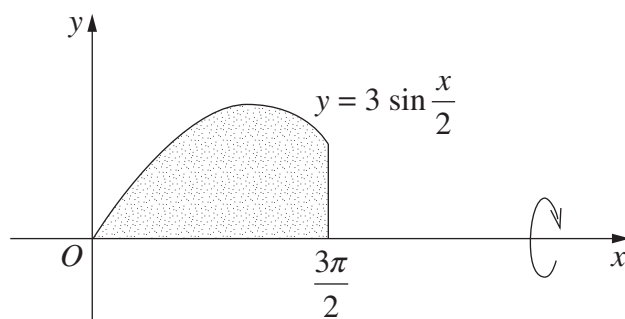
**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet.

(a) (i) Write  $\sqrt{3}\cos x - \sin x$  in the form  $2\cos(x + \alpha)$ , where  $0 < \alpha < \frac{\pi}{2}$ . **1**

(ii) Hence, or otherwise, solve  $\sqrt{3}\cos x = 1 + \sin x$ , where  $0 < x < 2\pi$ . **2**

- (b) The region bounded by the graph  $y = 3\sin\frac{x}{2}$  and the  $x$ -axis between  $x = 0$  and  $x = \frac{3\pi}{2}$  is rotated about the  $x$ -axis to form a solid. **3**



Find the exact volume of the solid.

- (c) A cup of coffee with an initial temperature of  $80^\circ\text{C}$  is placed in a room with a constant temperature of  $22^\circ\text{C}$ . **3**

The temperature,  $T^\circ\text{C}$ , of the coffee after  $t$  minutes is given by

$$T = A + Be^{-kt},$$

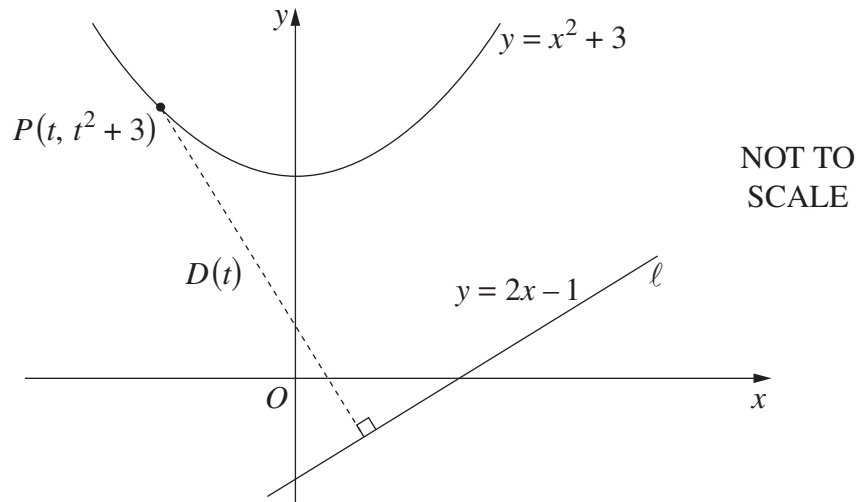
where  $A$ ,  $B$  and  $k$  are positive constants. The temperature of the coffee drops to  $60^\circ\text{C}$  after 10 minutes.

How long does it take for the temperature of the coffee to drop to  $40^\circ\text{C}$ ?  
Give your answer to the nearest minute.

**Question 12 continues on page 11**

Question 12 (continued)

- (d) The point  $P(t, t^2 + 3)$  lies on the curve  $y = x^2 + 3$ . The line  $\ell$  has equation  $y = 2x - 1$ . The perpendicular distance from  $P$  to the line  $\ell$  is  $D(t)$ .



- (i) Show that  $D(t) = \frac{t^2 - 2t + 4}{\sqrt{5}}$ . **2**
- (ii) Find the value of  $t$  when  $P$  is closest to  $\ell$ . **1**
- (iii) Show that, when  $P$  is closest to  $\ell$ , the tangent to the curve at  $P$  is parallel to  $\ell$ . **1**
- (e) A particle moves along a straight line. The displacement of the particle from the origin is  $x$ , and its velocity is  $v$ . The particle is moving so that  $v^2 + 9x^2 = k$ , where  $k$  is a constant. **2**

Show that the particle moves in simple harmonic motion with period  $\frac{2\pi}{3}$ .

**End of Question 12**

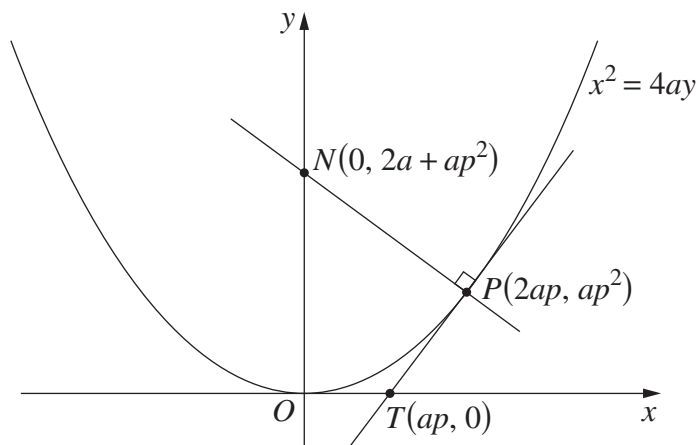
**Question 13** (15 marks) Use a SEPARATE writing booklet.

- (a) A spherical raindrop of radius  $r$  metres loses water through evaporation at a rate that depends on its surface area. The rate of change of the volume  $V$  of the raindrop is given by

$$\frac{dV}{dt} = -10^{-4} A,$$

where  $t$  is time in seconds and  $A$  is the surface area of the raindrop. The surface area and the volume of the raindrop are given by  $A = 4\pi r^2$  and  $V = \frac{4}{3}\pi r^3$  respectively.

- (i) Show that  $\frac{dr}{dt}$  is constant. **1**
  - (ii) How long does it take for a raindrop of volume  $10^{-6} \text{ m}^3$  to completely evaporate? **2**
- (b) The point  $P(2ap, ap^2)$  lies on the parabola  $x^2 = 4ay$ . The tangent to the parabola at  $P$  meets the  $x$ -axis at  $T(ap, 0)$ . The normal to the tangent at  $P$  meets the  $y$ -axis at  $N(0, 2a + ap^2)$ .



The point  $G$  divides  $NT$  externally in the ratio  $2 : 1$ .

- (i) Show that the coordinates of  $G$  are  $(2ap, -2a - ap^2)$ . **2**
- (ii) Show that  $G$  lies on a parabola with the same directrix and focal length as the original parabola. **2**

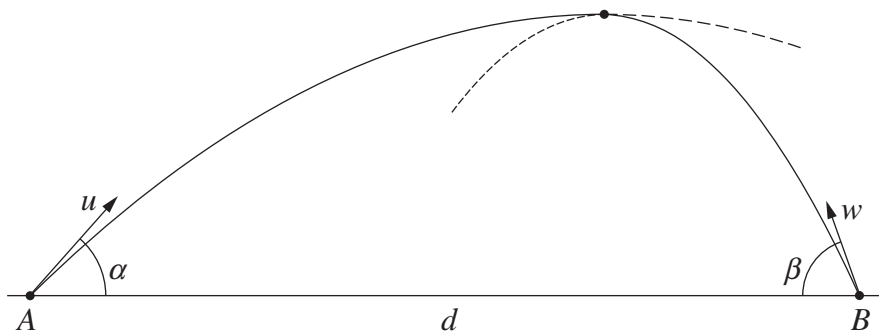
**Question 13 continues on page 13**

Question 13 (continued)

- (c) Points  $A$  and  $B$  are located  $d$  metres apart on a horizontal plane. A projectile is fired from  $A$  towards  $B$  with initial velocity  $u \text{ m s}^{-1}$  at angle  $\alpha$  to the horizontal.

At the same time, another projectile is fired from  $B$  towards  $A$  with initial velocity  $w \text{ m s}^{-1}$  at angle  $\beta$  to the horizontal, as shown on the diagram.

The projectiles collide when they both reach their maximum height.



The equations of motion of a projectile fired from the origin with initial velocity  $V \text{ m s}^{-1}$  at angle  $\theta$  to the horizontal are

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{g}{2} t^2. \quad (\text{Do NOT prove this.})$$

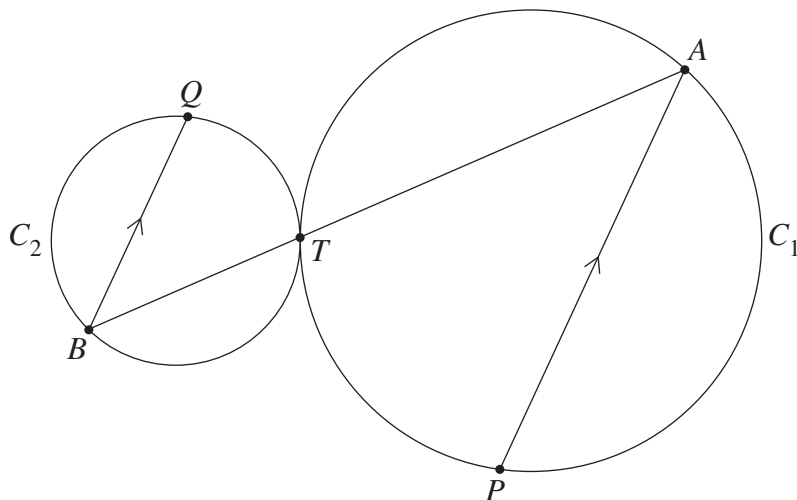
- |                                                                                   |          |
|-----------------------------------------------------------------------------------|----------|
| (i) How long does the projectile fired from $A$ take to reach its maximum height? | <b>2</b> |
| (ii) Show that $u \sin \alpha = w \sin \beta$ .                                   | <b>1</b> |
| (iii) Show that $d = \frac{uw}{g} \sin(\alpha + \beta)$ .                         | <b>2</b> |

**Question 13 continues on page 14**

Question 13 (continued)

- (d) The circles  $C_1$  and  $C_2$  touch at the point  $T$ . The points  $A$  and  $P$  are on  $C_1$ . The line  $AT$  intersects  $C_2$  at  $B$ . The point  $Q$  on  $C_2$  is chosen so that  $BQ$  is parallel to  $PA$ .

3



Copy or trace the diagram into your writing booklet.

Prove that the points  $Q$ ,  $T$  and  $P$  are collinear.

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that for  $k > 0$ ,  $\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$ . **1**

(ii) Use mathematical induction to prove that for all integers  $n \geq 2$ , **3**

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}.$$

(b) (i) Write down the coefficient of  $x^{2n}$  in the binomial expansion of  $(1+x)^{4n}$ . **1**

(ii) Show that  $(1+x^2+2x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^{2n-k} (x+2)^{2n-k}$ . **2**

(iii) It is known that **3**

$$\begin{aligned} x^{2n-k} (x+2)^{2n-k} &= \binom{2n-k}{0} 2^{2n-k} x^{2n-k} + \binom{2n-k}{1} 2^{2n-k-1} x^{2n-k+1} \\ &\quad + \cdots + \binom{2n-k}{2n-k} 2^0 x^{4n-2k}. \quad (\text{Do NOT prove this.}) \end{aligned}$$

Show that

$$\binom{4n}{2n} = \sum_{k=0}^n 2^{2n-2k} \binom{2n}{k} \binom{2n-k}{k}.$$

(c) The equation  $e^t = \frac{1}{t}$  has an approximate solution  $t_0 = 0.5$ .

(i) Use one application of Newton's method to show that  $t_1 = 0.56$  is another approximate solution of  $e^t = \frac{1}{t}$ . **2**

(ii) Hence, or otherwise, find an approximation to the value of  $r$  for which the graphs  $y = e^{rx}$  and  $y = \log_e x$  have a common tangent at their point of intersection. **3**

**End of paper**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$





## **2013 HSC Mathematics Extension 1 Marking Guidelines**

### **Section I**

#### **Multiple-choice Answer Key**

| <b>Question</b> | <b>Answer</b> |
|-----------------|---------------|
| 1               | C             |
| 2               | D             |
| 3               | C             |
| 4               | D             |
| 5               | A             |
| 6               | B             |
| 7               | A             |
| 8               | D             |
| 9               | B             |
| 10              | C             |

## Section II

### Question 11 (a)

| Criteria         | Marks |
|------------------|-------|
| • Correct answer | 1     |

For any cubic  $\alpha\beta\gamma = -\frac{d}{a}$ , so

$$\alpha\beta\gamma = -\frac{7}{2}.$$

### Question 11 (b)

| Criteria                                                 | Marks |
|----------------------------------------------------------|-------|
| • Correct primitive                                      | 2     |
| • Attempts to use standard integral, or equivalent merit | 1     |

$$\begin{aligned}
 & \int \frac{1}{\sqrt{49 - 4x^2}} dx \\
 &= \int \frac{1}{\sqrt{4\left(\frac{49}{4} - x^2\right)}} dx \\
 &= \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{7}{2}\right)^2 - x^2}} dx \\
 &= \frac{1}{2} \sin^{-1} \frac{x}{\frac{7}{2}} + c \\
 &= \frac{1}{2} \sin^{-1} \frac{2x}{7} + c
 \end{aligned}$$

**Question 11 (c)**

| Criteria                                                       | Marks |
|----------------------------------------------------------------|-------|
| • Correct answer                                               | 2     |
| • Identifies correct binomial coefficient, or equivalent merit | 1     |

The probability is

$$\begin{aligned}
 {}^{10}C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^3 &= \frac{10!}{7!3!} \frac{3^3}{4^{10}} \\
 &= 120 \cdot \frac{3^3}{4^{10}} \\
 &= 30 \cdot \frac{3^3}{4^9}.
 \end{aligned}$$

For each question, the probability of choosing the correct option is  $\frac{1}{4}$  and the probability of choosing an incorrect option is  $\frac{3}{4}$ .  
The probability of choosing the correct option for exactly 7 out of 10 questions is the term in  $\left(\frac{1}{4} + \frac{3}{4}\right)^{10}$  containing  $\left(\frac{1}{4}\right)^7$ .

**Question 11 (d) (i)**

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 2     |
| • Finds $f'(x)$    | 1     |

Domain: all real  $x \neq \pm 2$

Using the quotient rule

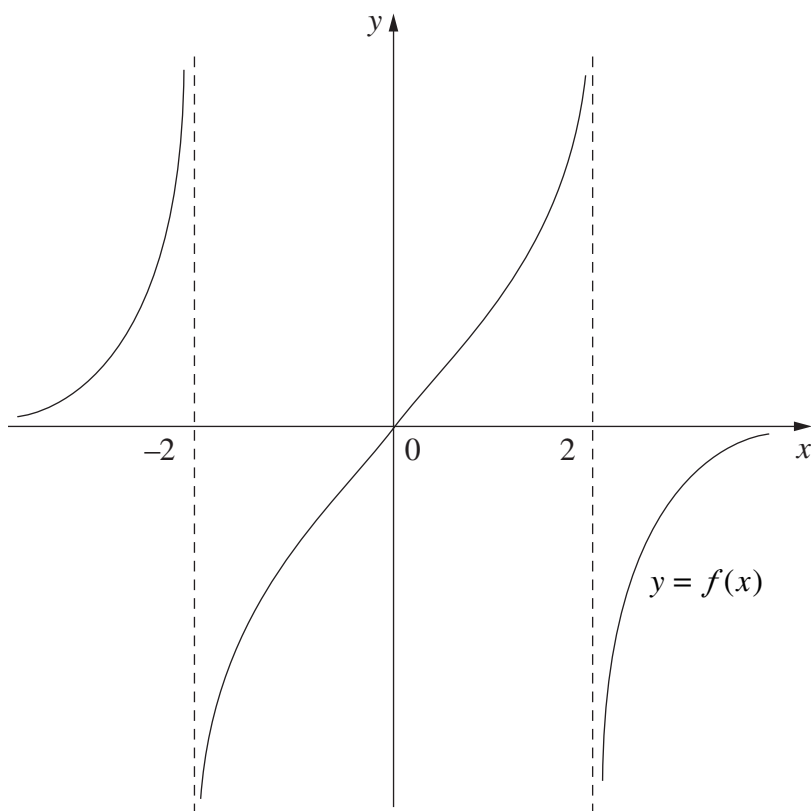
$$\begin{aligned}
 f'(x) &= \frac{(4 - x^2) \cdot 1 - x(-2x)}{(4 - x^2)^2} \\
 &= \frac{4 - x^2 + 2x^2}{(4 - x^2)^2} \\
 &= \frac{4 + x^2}{(4 - x^2)^2}
 \end{aligned}$$

$f'(x) > 0$  since  $4 + x^2 > 0$  for all  $x$ , and  $(4 - x^2)^2 > 0$  for all  $x \neq \pm 2$

Hence  $f'(x) > 0$  for all  $x \neq \pm 2$ .

**Question 11 (d) (ii)**

| Criteria                                              | Marks |
|-------------------------------------------------------|-------|
| • Correct graph                                       | 2     |
| • Finds both vertical asymptotes, or equivalent merit | 1     |


**Question 11 (e)**

| Criteria         | Marks |
|------------------|-------|
| • Correct answer | 1     |

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{3x} &= \lim_{x \rightarrow 0} \frac{1}{6} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \\
 &= \frac{1}{6} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \\
 &= \frac{1}{6} \left( \text{since } \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1. \right)
 \end{aligned}$$

**Question 11 (f)**

| Criteria                                           | Marks |
|----------------------------------------------------|-------|
| • Correct solution                                 | 3     |
| • Finds a primitive of the form of $a \tan^{-1} u$ | 2     |
| • Attempts to use given substitution               | 1     |

$$u = e^{3x}, \quad \frac{du}{dx} = 3e^{3x}, \quad u^2 = e^{6x}$$

If  $x = 0$ , then  $u = e^0 = 1$

If  $x = \frac{1}{3}$ , then  $u = e^{3 \cdot \frac{1}{3}} = e$

Hence

$$\begin{aligned} \int_0^{\frac{1}{3}} \frac{e^{3x}}{e^{6x} + 1} dx \\ &= \frac{1}{3} \int_1^e \frac{1}{u^2 + 1} du \\ &= \frac{1}{3} \left[ \tan^{-1} u \right]_1^e \\ &= \frac{1}{3} \left( \tan^{-1} e - \tan^{-1} 1 \right) \\ &= \frac{1}{3} \left( \tan^{-1} e - \frac{\pi}{4} \right). \end{aligned}$$

**Question 11 (g)**

| Criteria                                                          | Marks |
|-------------------------------------------------------------------|-------|
| • Correct solution                                                | 2     |
| • Uses correct derivative of $\sin^{-1} 5x$ , or equivalent merit | 1     |

Using the product rule

$$\begin{aligned} f'(x) &= x^2 \frac{5}{\sqrt{1-25x^2}} + 2x \sin^{-1} 5x \\ &= \frac{5x^2}{\sqrt{1-25x^2}} + 2x \sin^{-1} 5x. \end{aligned}$$

**Question 12 (a) (i)**

| Criteria         | Marks |
|------------------|-------|
| • Correct answer | 1     |

We want

$$\begin{aligned}\sqrt{3}\cos x - \sin x &= 2\cos(x + \alpha) \\ \frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x &= \cos(x + \alpha) \\ &= \cos x \cos \alpha - \sin x \sin \alpha, \\ \text{So } \cos \alpha &= \frac{\sqrt{3}}{2} \text{ and } \sin \alpha = \frac{1}{2}. \text{ Hence } \tan \alpha = \frac{1}{\sqrt{3}}. \\ \therefore \alpha &= \frac{\pi}{6} \left( \text{and } 0 < \frac{\pi}{6} < \frac{\pi}{2} \right)\end{aligned}$$

**Question 12 (a) (ii)**

| Criteria                                                                       | Marks |
|--------------------------------------------------------------------------------|-------|
| • Correct solution                                                             | 2     |
| • Obtains an equation of the form $\cos(\text{---}) = k$ , or equivalent merit | 1     |

Rearrange the equation and use part (i):

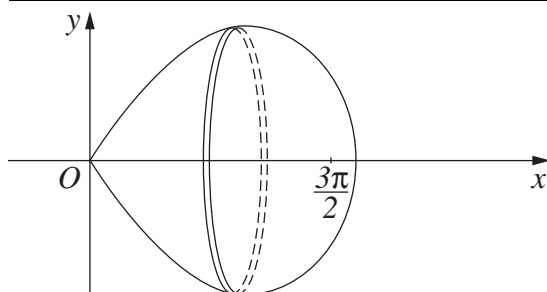
$$\begin{aligned}\sqrt{3}\cos x - \sin x &= 1, \text{ ie } 2\cos\left(x + \frac{\pi}{6}\right) = 1 \\ \cos\left(x + \frac{\pi}{6}\right) &= \frac{1}{2} \\ \therefore x + \frac{\pi}{6} &= \frac{\pi}{3} \text{ or } x + \frac{\pi}{6} = \frac{5\pi}{3}\end{aligned}$$

Hence the solutions satisfying  $0 < x < 2\pi$  are

$$\begin{aligned}x &= \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6} \\ \text{and } x &= \frac{5\pi}{3} - \frac{\pi}{6} = \frac{3\pi}{2}.\end{aligned}$$

### Question 12 (b)

| Criteria                                       | Marks |
|------------------------------------------------|-------|
| • Correct solution                             | 3     |
| • Finds correct primitive, or equivalent merit | 2     |
| • Finds correct integrand, or equivalent merit | 1     |



Volume of solid is given by

$$V = \pi \int y^2 dx$$

$$V = \pi \int_0^{\frac{3\pi}{2}} \left( 3 \sin \frac{x}{2} \right)^2 dx$$

$$= 9\pi \int_0^{\frac{3\pi}{2}} \sin^2 \frac{x}{2} dx$$

$$V = 9\pi \int_0^{\frac{3\pi}{2}} \frac{1}{2} (1 - \cos x) dx \quad \left( \begin{array}{l} \text{Note: } \cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} \\ \\ = 1 - 2 \sin^2 \frac{x}{2} \\ \\ \therefore \sin^2 \frac{x}{2} = \frac{1}{2} (1 - \cos x) \end{array} \right)$$

$$= \frac{9\pi}{2} \left[ (x - \sin x) \right]_0^{\frac{3\pi}{2}}$$

$$= \frac{9\pi}{2} \left[ \left( \frac{3\pi}{2} - \sin \frac{3\pi}{2} \right) - (0 - \sin 0) \right]$$

$$= \frac{9\pi}{2} \left( \frac{3\pi}{2} + 1 \right)$$

The volume is  $\frac{9\pi}{2} \left( \frac{3\pi}{2} + 1 \right)$  units<sup>3</sup>.

### Question 12 (c)

| Criteria                                                   | Marks |
|------------------------------------------------------------|-------|
| • Correct solution                                         | 3     |
| • Finds values for $A$ , $B$ and $k$ , or equivalent merit | 2     |
| • Finds value for $A$ , or equivalent merit                | 1     |

As  $t \rightarrow \infty$ ,  $T$  approaches the temperature of the room, and so  $A = 22$ .

When  $t = 0$ ,  $T = 22 + B$

$$= 80$$

$$\therefore B = 58.$$

Hence

$$T = 22 + 58e^{-kt}$$

To find  $k$ : After 10 minutes the temperature is  $60^\circ$  ( $t = 10$ ,  $T = 60$ )

$$60 = 22 + 58e^{-k \times 10}$$

$$38 = 58e^{-10k}$$

$$\log_e \left( \frac{38}{58} \right) = -10k,$$

$$\text{so } k = \frac{-1}{10} \log_e \left( \frac{38}{58} \right)$$

To determine time to drop to  $40^\circ\text{C}$ :

$$40 = 22 + 58e^{\frac{1}{10} \log_e \left( \frac{38}{58} \right) t}$$

$$\frac{18}{58} = e^{\frac{1}{10} \log_e \left( \frac{38}{58} \right) t}$$

$$\log_e \frac{18}{58} = \frac{1}{10} \log_e \left( \frac{38}{58} \right) t$$

$$t = 10 \frac{\log_e \left( \frac{18}{58} \right)}{\log_e \left( \frac{38}{58} \right)},$$

$$t = 27.67...$$

$\therefore$  temperature drops to  $40^\circ$  after 28 minutes (to the nearest minute).



**Question 12 (d) (i)**

| Criteria                                               | Marks |
|--------------------------------------------------------|-------|
| • Correct solution                                     | 2     |
| • Uses an appropriate method to attempt to find $D(t)$ | 1     |

Using the perpendicular distance formula from  $P(t, t^2 + 3)$  to the line  $y = 2x - 1$ :

$$\begin{aligned}
 D(t) &= \frac{|2t - (t^2 + 3) - 1|}{\sqrt{4 + 1}} \\
 &= \frac{|-t^2 + 2t - 4|}{\sqrt{5}} \\
 &= \frac{|-(t^2 - 2t + 4)|}{\sqrt{5}} \\
 &= \frac{|t^2 - 2t + 4|}{\sqrt{5}}
 \end{aligned}$$

Now  $t^2 - 2t + 4 = (t - 1)^2 + 3 > 0$  for all  $t$ , so

$$D(t) = \frac{t^2 - 2t + 4}{\sqrt{5}} \text{ is the perpendicular distance.}$$

**Question 12 (d) (ii)**

| Criteria         | Marks |
|------------------|-------|
| • Correct answer | 1     |

To find minimum distance,  $D'(t) = 0$ .

$$\begin{aligned}
 D'(t) &= \frac{1}{\sqrt{5}}(2t - 2) \\
 &= \frac{2}{\sqrt{5}}(t - 1) \\
 &= 0
 \end{aligned}$$

$$\therefore t = 1$$

Minimum because  $D(t) \rightarrow \infty$  as  $t \rightarrow \pm\infty$ .

Alternative:  $D(t)$  is a quadratic with minimum at  $t = 1$  since  $D(t) = \frac{1}{\sqrt{5}}((t - 1)^2 + 3)$ .

**Question 12 (d) (iii)**

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

$t = 1$  corresponds to  $x = 1$ , on the curve given by  $y = x^2 + 3$ .

$y' = 2x$ , so the slope of the tangent at  $x = 1$  is 2.

Also, the slope of the line  $y = 2x - 1$  is 2.

$\therefore$  the line and the tangent are parallel, since their gradients are equal.

**Question 12 (e)**

| Criteria                                                                   | Marks |
|----------------------------------------------------------------------------|-------|
| • Correct solution                                                         | 2     |
| • Correctly differentiates $v^2$ with respect to $x$ , or equivalent merit | 1     |

$$v^2 + 9x^2 = k$$

$$v^2 = k - 9x^2$$

$$\frac{1}{2}v^2 = \frac{1}{2}k - \frac{9}{2}x^2$$

$$\ddot{x} = \frac{d}{dx} \left( \frac{1}{2}v^2 \right) = -9x$$

For SHM,  $\ddot{x} = -n^2(x - b)$ .

Now,  $\ddot{x} = -9x$ , is of this form with

$$n = 3 \text{ and } b = 0$$

As  $n = 3$ , the period is  $\frac{2\pi}{3}$ .

**Question 13 (a) (i)**

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

Using  $V = \frac{4}{3}\pi r^3$  and  $A = 4\pi r^2$ ,

$$\begin{aligned}
 -10^{-4}A &= \frac{dV}{dt} \\
 &= \frac{dV}{dr} \frac{dr}{dt} \\
 &= 4\pi r^2 \frac{dr}{dt} \\
 &= A \frac{dr}{dt}
 \end{aligned}$$

Hence  $\frac{dr}{dt} = -10^{-4}$ , which is constant.

**Question 13 (a) (ii)**

| Criteria                                    | Marks |
|---------------------------------------------|-------|
| • Correct answer                            | 2     |
| • Finds initial radius, or equivalent merit | 1     |

From part (i)  $\frac{dr}{dt} = 10^{-4}$

To determine the initial radius of the raindrop:

$$\frac{4}{3}\pi r^3 = 10^{-6},$$

$$\text{so } r^3 = \frac{3}{4\pi} 10^{-6}$$

$$r = \sqrt[3]{\frac{3}{4\pi}} 10^{-2}$$

$$= \frac{1}{10^2} \sqrt[3]{\frac{3}{4\pi}}$$

Time to evaporate;

$$\begin{aligned}
 t &= \frac{r}{10^{-4}} \\
 &= 10^2 \sqrt[3]{\frac{3}{4\pi}}
 \end{aligned}$$

$\therefore$  time = 62 seconds (to the nearest second).

**Question 13 (b) (i)**

| Criteria                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------|-------|
| • Correct solution                                                                                 | 2     |
| • Writes an unsimplified expression for the $x$ - or $y$ - coordinate of $G$ , or equivalent merit | 1     |

The point  $G$  divides  $NT$  in the ratio 2:1 externally means that  $T$  is the midpoint of  $NG$ .

Hence if  $(x_0, y_0)$  are the coordinates of  $G$ , then

$$\frac{0 + x_0}{2} = ap \quad (x\text{-coordinate of } T)$$

$$\frac{2a + ap^2 + y_0}{2} = 0 \quad (y\text{-coordinate of } T)$$

$$\therefore x_0 = 2ap \text{ and } y_0 = -2a - ap^2$$

ie  $G$  is the point  $(2ap, -2a - ap^2)$ , as required

### Question 13 (b) (ii)

| Criteria                                                 | Marks |
|----------------------------------------------------------|-------|
| • Correct solution for $G$                               | 2     |
| • Shows that $G$ lies on a parabola, or equivalent merit | 1     |

The original parabola has focal length  $a$  and directrix  $y = -a$ . To show that  $G(2ap, -2a - ap^2)$  lies on a parabola, eliminate  $p$  from  $x = 2ap$ ,  $y = -2a - ap^2$

Now,  $p = \frac{x}{2a}$ , so

$$y = -2a - a\left(\frac{x}{2a}\right)^2$$

$$y = -2a - \frac{ax^2}{4a^2}$$

$$= -2a - \frac{x^2}{4a}$$

$$y + 2a = -\frac{x^2}{4a}$$

$$x^2 = -4a(y + 2a)$$

This is a parabola with focal length  $a$  and vertex  $(0, -2a)$ .

So the directrix is

$$\begin{aligned} y &= -2a + a \\ &= -a \end{aligned}$$

Hence focal length and directrix are the same for both parabolas.

Alternative:

$(2ap, -ap^2)$  is the parabola obtained if the original parabola is reflected about the  $x$ -axis. It has focal length  $a$  and directrix  $y = +a$ .

Now  $G(2ap, -2a - ap^2)$  is obtained by translating the reflected parabola down by  $2a$  units. Translation preserves the focal length  $a$ , and the directrix becomes  $y = a - 2a = -a$ .

**Question 13 (c) (i)**

| Criteria                                              | Marks |
|-------------------------------------------------------|-------|
| • Correct solution                                    | 2     |
| • Obtains an expression for $y$ , or equivalent merit | 1     |

We need to find the time when the quadratic  $y = ut \sin \alpha - \frac{g}{2}t^2$  is a maximum.

$$\frac{dy}{dt} = u \sin \alpha - gt$$

$$= 0 \text{ for max/min}$$

$$t = \frac{u \sin \alpha}{g}$$

This will be at the maximum height since  $y = ut \sin \alpha - \frac{g}{2}t^2$  is a concave down parabola.

**Question 13 (c) (ii)**

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

Similarly from part (i), the projectile fired from  $B$  reaches its maximum height when

$$t = \frac{w \sin \beta}{g}.$$

The projectiles collide when they both reach their maximum height.

$$\text{ie } \frac{u \sin \alpha}{g} = t = \frac{w \sin \beta}{g},$$

$$\text{so } u \sin \alpha = w \sin \beta.$$

**Question 13 (c) (iii)**

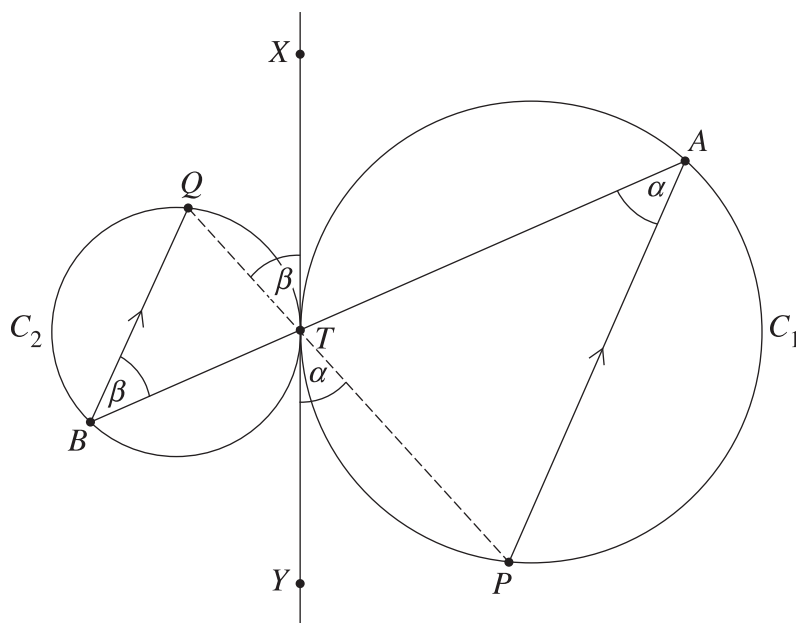
| Criteria                                          | Marks |
|---------------------------------------------------|-------|
| • Correct solution                                | 2     |
| • Finds the distance one projectile has travelled | 1     |

When the projectiles collide, the sum of the two distances travelled must be  $d$ . Both projectiles have travelled for time  $t$ .

$$\begin{aligned}
 t &= \frac{u \sin \alpha}{g} = \frac{w \sin \beta}{g}, \text{ so} \\
 d &= u \frac{u \sin \alpha}{g} \cos \alpha + w \frac{w \sin \beta}{g} \cos \beta \\
 &= u \frac{w \sin \beta}{g} \cos \alpha + w \frac{u \sin \alpha}{g} \cos \beta \\
 &= \frac{uw}{g} (\sin \beta \cos \alpha + \sin \alpha \cos \beta) \\
 &= \frac{uw}{g} \sin(\alpha + \beta).
 \end{aligned}$$

### Question 13 (d)

| Criteria                                                                                                                 | Marks |
|--------------------------------------------------------------------------------------------------------------------------|-------|
| • Correct solution                                                                                                       | 3     |
| • Proves that $\angle QBT$ and $\angle PAT$ are equal AND uses the properties of the common tangent, or equivalent merit | 2     |
| • Proves that $\angle QBT$ and $\angle PAT$ are equal or uses the properties of the common tangent, or equivalent merit  | 1     |



The circles  $C_1$  and  $C_2$  have a common tangent at  $T$ .

Choose points  $X$  and  $Y$  on that tangent as marked on the diagram above.

$$\angle YTP = \angle PAT = \alpha \quad (\text{angle in alternate segment, chord } PT \text{ and tangent } XY)$$

$$\angle XTQ = \angle QBT = \beta \quad (\text{angle in alternate segment, chord } QT \text{ and tangent } XY)$$

$$\text{But } \angle PAT = \angle QBT \quad (\text{alternate angles equal, } BQ \parallel PA)$$

$$\text{ie } \alpha = \beta$$

$$\therefore \angle YTP = \angle XTQ$$

Hence  $QP$  is a straight line making equal vertically opposite angles with line  $XY$  at  $T$ .

$\therefore P, T, Q$  are collinear.



**Question 14 (a) (i)**

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

$$\begin{aligned}\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} &= \frac{1}{(k+1)^2} - \frac{k+1}{k(k+1)} + \frac{k}{k(k+1)} \\ &= \frac{1}{k+1} \left[ \frac{1}{k+1} - \frac{1}{k} \right] \\ &< 0 \quad (\text{since } k+1 > k \text{ and } k > 0).\end{aligned}$$

**Question 14 (a) (ii)**

| Criteria                                                                        | Marks |
|---------------------------------------------------------------------------------|-------|
| • Correct solution                                                              | 3     |
| • Uses correct assumption in an attempt to prove that $p(k) \Rightarrow p(k+1)$ | 2     |
| • Establishes initial case, or equivalent merit                                 | 1     |

Case  $n = 2$ :  $LHS \frac{1}{1} + \frac{1}{2^2} = 1 + \frac{1}{4} = \frac{5}{4}$   $RHS: 2 - \frac{1}{2} = \frac{3}{2}$

$$LHS < RHS$$

Hence true for  $n = 2$

Assume true for some  $k \geq 2$

ie  $\sum_{p=1}^k \frac{1}{p^2} < 2 - \frac{1}{k}$

Need to show true for  $n = k+1$

ie  $\sum_{p=1}^{k+1} \frac{1}{p^2} < 2 - \frac{1}{k+1}$

Now

$$\begin{aligned}LHS &= \sum_{p=1}^{k+1} \frac{1}{p^2} = \sum_{p=1}^k \frac{1}{p^2} + \frac{1}{(k+1)^2} \\ &< 2 - \frac{1}{k} + \frac{1}{(k+1)^2} \quad (\text{by induction assumption}) \\ &< 2 - \frac{1}{k} + \frac{1}{k} - \frac{1}{k+1} \quad \left( \text{using part (i), ie } \frac{1}{(k+1)^2} < \frac{1}{k} - \frac{1}{k+1} \right) \\ \text{ie } \sum_{p=1}^{k+1} \frac{1}{p^2} &< 2 - \frac{1}{k+1}\end{aligned}$$

By the principle of Mathematical Induction, it is true for all  $n \geq 2$ .

**Question 14 (b) (i)**

| Criteria         | Marks |
|------------------|-------|
| • Correct answer | 1     |

From the binomial theorem

$$(1+x)^{4n} = \sum_{k=0}^{4n} \binom{4n}{k} x^{4n-k}$$

The coefficient of  $x^{2n}$  is  $\binom{4n}{2n}$ , choosing  $k = 2n$ .

**Question 14 (b) (ii)**

| Criteria                                                                                            | Marks |
|-----------------------------------------------------------------------------------------------------|-------|
| • Correct solution                                                                                  | 2     |
| • Writes $(1+x^2+2x)^{2n}$ as $(1+x(x+2))^{2n}$ or uses the binomial expansion, or equivalent merit | 1     |

From the binomial theorem

$$\begin{aligned} (1+x^2+2x)^{2n} &= (1+x(x+2))^{2n} \\ &= \sum_{k=0}^{2n} \binom{2n}{k} (x(x+2))^{2n-k} \\ &= \sum_{k=0}^{2n} \binom{2n}{k} x^{2n-k} (x+2)^{2n-k}. \end{aligned}$$

**Question 14 (b) (iii)**

| Criteria                                                                                  | Marks |
|-------------------------------------------------------------------------------------------|-------|
| • Correct solution                                                                        | 3     |
| • Substitutes the given expression into part (ii) and makes progress towards the solution | 2     |
| • Connects $(1+x)^{4n}$ and $(1+x^2+2x)^{2n}$ , or equivalent merit                       | 1     |

$$(1+x^2+2x) = (1+x)^2, \text{ so}$$

$$(1+x^2+2x)^{2n} = (1+x)^{4n}$$

By part (i),  $\binom{4n}{2n}$  is the coefficient of  $x^{2n}$  in  $(1+x^2+2x)^{2n}$ .

From part (ii) these are

the terms in  $\binom{2n}{k} x^{2n-k} (x+2)^{2n-k}$  for  $k = 0, \dots, n$ . If  $k > n$  there is no term involving  $x^{2n}$ .

Using the given identity the coefficients of  $x^{2n}$  are:

$$k = 0 : \binom{2n}{0} \binom{2n-0}{0} 2^{2n-0}$$

$$k = 1 : \binom{2n}{1} \binom{2n-1}{1} 2^{2n-1-1}$$

$$k = 2 : \binom{2n}{2} \binom{2n-2}{2} 2^{2n-2-2}$$

$\vdots$

$$k = n : \binom{2n}{n} \binom{2n-n}{n} 2^{2n-n-n}$$

Hence

$$\begin{aligned} \binom{4n}{2n} &= \binom{2n}{0} \binom{2n}{0} 2^{2n} + \binom{2n}{1} \binom{2n-1}{1} 2^{2n-2} + \dots + \binom{2n}{n} \binom{n}{n} 2^0 \\ &= \sum_{k=0}^n \binom{2n}{k} \binom{2n-k}{k} 2^{2n-2k}. \end{aligned}$$

**Question 14 (c) (i)**

| Criteria                            | Marks |
|-------------------------------------|-------|
| • Correct solution                  | 2     |
| • Correctly applies Newton's method | 1     |

Take  $f(t) = e^t - \frac{1}{t}$

Then  $f'(t) = e^t + \frac{1}{t^2}$

Apply Newton's method

$$t_1 = t_0 - \frac{f(t_0)}{f'(t_0)}$$

$$= 0.5 - \frac{e^{0.5} - 2}{e^{0.5} + 4}$$

$$= 0.5621\dots$$

Hence  $t_1 = 0.56$  is another approximate solution.

**Question 14 (c) (ii)**

| Criteria                                                                                             | Marks |
|------------------------------------------------------------------------------------------------------|-------|
| • Correct solution                                                                                   | 3     |
| • Uses part (c) (i) to find an approximate solution to $re^{rx} = \frac{1}{x}$ , or equivalent merit | 2     |
| • Establishes $re^{rx} = \frac{1}{x}$ , or equivalent merit                                          | 1     |

We need to determine  $r$  and  $x$  so that

$$e^{rx} = \log_e x \text{ and } re^{rx} = \frac{1}{x}$$

(common point and equal slope of tangents).

From the equal slopes

$$e^{rx} = \frac{1}{rx}$$

Setting  $rx = t$  and using part (i), we have an approximate solution

$$rx = 0.56$$

Since the curves intersect at  $x$ ,

$$\log_e x = e^{rx} = e^{0.56}$$

$$x = e^{e^{0.56}} = 5.758\dots$$

$$\text{now } rx = 0.56$$

$$\therefore r = \frac{0.56}{5.758\dots}$$

$$= 0.0972$$

$$r \doteq 0.097$$

# Mathematics Extension 1

## 2013 HSC Examination Mapping Grid

### Section I

| Question | Marks | Content | Syllabus outcomes |
|----------|-------|---------|-------------------|
| 1        | 1     | 16.2    | PE3               |
| 2        | 1     | 15.1    | HE4               |
| 3        | 1     | 2.8     | PE3               |
| 4        | 1     | 16.1    | PE3               |
| 5        | 1     | 11.5    | HE6               |
| 6        | 1     | 15.2    | HE4               |
| 7        | 1     | 18.1    | H5, PE3           |
| 8        | 1     | 5.7     | H5, PE2           |
| 9        | 1     | 15.3    | HE4               |
| 10       | 1     | 1.4E    | PE3               |

### Section II

| Question     | Marks | Content     | Syllabus outcomes |
|--------------|-------|-------------|-------------------|
| 11 (a)       | 1     | 16.3        | PE3               |
| 11 (b)       | 2     | 15.5        | HE4               |
| 11 (c)       | 2     | 18.2        | HE3               |
| 11 (d) (i)   | 2     | 8.8         | P7, HE4           |
| 11 (d) (ii)  | 2     | 10.5        | H6, HE4           |
| 11 (e)       | 1     | 13.4        | H5, HE7           |
| 11 (f)       | 3     | 11.5, 15.5  | HE4, HE6          |
| 11 (g)       | 2     | 8.8, 15.5   | P7, HE4           |
| 12 (a) (i)   | 1     | 5.7, 5.9    | PE2               |
| 12 (a) (ii)  | 2     | 5.7, 5.9    | PE2               |
| 12 (b)       | 3     | 11.4, 13.6E | H5, H8, HE7       |
| 12 (c)       | 3     | 14.2E       | HE3               |
| 12 (d) (i)   | 2     | 6.5         | P4                |
| 12 (d) (ii)  | 1     | 8.5, 10.6   | P4, H5            |
| 12 (d) (iii) | 1     | 8.5         | P6                |
| 12 (e)       | 2     | 14.4        | HE3               |
| 13 (a) (i)   | 1     | 14.1        | HE5               |
| 13 (a) (ii)  | 2     | 14.1        | HE5               |

| Question     | Marks | Content               | Syllabus outcomes |
|--------------|-------|-----------------------|-------------------|
| 13 (b) (i)   | 2     | 6.7E, 9.6             | PE3               |
| 13 (b) (ii)  | 2     | 9.6                   | PE3, PE4          |
| 13 (c) (i)   | 2     | 14.3                  | HE3               |
| 13 (c) (ii)  | 1     | 14.3                  | HE3               |
| 13 (c) (iii) | 2     | 5.7, 14.3             | PE2, HE3          |
| 13 (d)       | 3     | 2.3, 2.9              | P4, PE2, PE3, PE6 |
| 14 (a) (i)   | 1     | 1.4E                  | PE3               |
| 14 (a) (ii)  | 3     | 7.4                   | HE2               |
| 14 (b) (i)   | 1     | 17.3                  | HE3               |
| 14 (b) (ii)  | 2     | 17.3                  | HE3               |
| 14 (b) (iii) | 3     | 17.3                  | HE3               |
| 14 (c) (i)   | 2     | 16.4                  | HE7               |
| 14 (c) (ii)  | 3     | 8.6, 12.2, 12.4, 12.5 | P6, H3, HE7       |